

M1 — Affinity.lean

Hellinger affinity, total in $\mathbb{R}_{\geq 0}^{\infty}$, symmetric;

$$H(\mu, \nu) = 0 \iff \mu \perp_m \nu$$



M2 — FiniteProduct.lean

pi_withDensity, integral product Fubini,
affinity tensorization over finite products



M3 — Cylinder.lean

cylinder density identity, affinity moments,
infinite-product / summability bridge



M4 — Singular.lean

singular direction:
cylinder squeeze + Borel–Cantelli

M5 — Equivalence.lean

absolutely continuous direction:
quantitative cylinder comparison



M6 — PoissonSingular.lean, PoissonDichotomy.lean

the application: poissonProduct_dichotomy —
product Poisson vs. product Haar, phase boundary at $\sigma = 1/2$